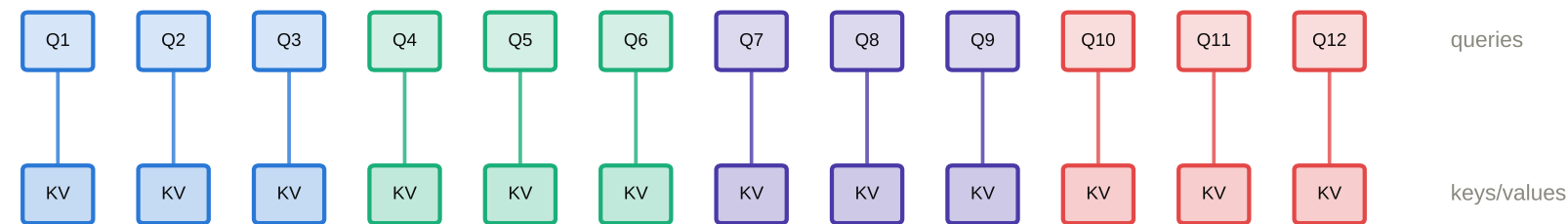


Grouped-Query Attention (GQA) vs Multi-Head Attention (MHA)

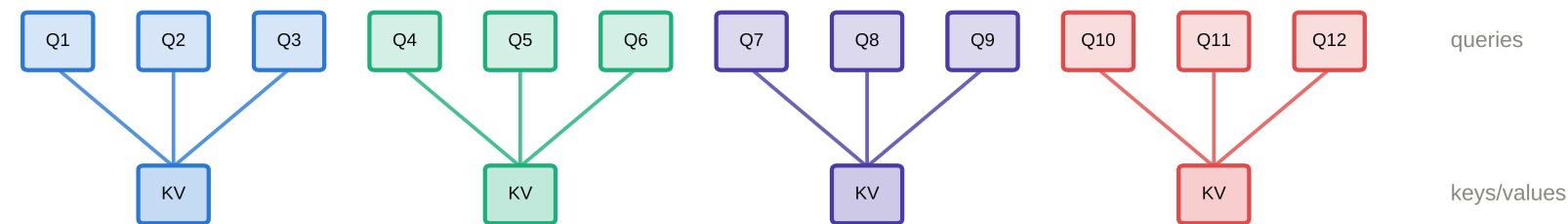
MHA — 12 query · 12 KV heads

every query head has its own K/V → largest KV cache



GQA — 12 query · 4 KV heads (this model)

each K/V head is shared by 3 query heads → 3× smaller KV cache



GQA interpolates between MHA ($n_{kv} = n_{head}$, full quality, full cache) and MQA ($n_{kv} = 1$, tiny cache, some quality loss). With $n_{kv} = 4$ the per-token KV cache is 3× smaller than MHA — cheaper long-context inference — at nearly no quality cost, because attention quality is dominated by the number of QUERY heads, which stays 12.